Nikhil Rajpoot

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.29
JAWAHAR NAVODAYA VIDYALAYA BARUASAGAR JHANSI UP	Class XII (CBSE)	2022	72.20%
JAWAHAR NAVODAYA VIDYALAYA BARUASAGAR JHANSI UP	Class X (CBSE)	2020	84.80%

SCHOLASTIC ACHIEVEMENTS

- Solved **350+** Data Structures and Algorithms problems in C++.
- **Dakshana Scholar (2022)** - Selected among **300 JEE aspirants across India** through **JDST** for a **fully funded, one-year residential JEE program.**
- **Cleared JNVST 2015**, securing admission to the **only Jawahar Navodaya Vidyalaya in the district** through a competitive selection process for free residential education from Class 6 to 12.

SKILLS

- **Languages:** C++, Python, SQL
- **Frameworks/Libraries:** React.js, NumPy, Pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow, LangChain, LangGraph
- **Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems (DBMS), Machine Learning, Large Language Models, Retrieval-Augmented Generation, Model Context Protocol
- **Databases:** MySQL, PostgreSQL
- **Tools:** Git, GitHub, Jupyter Notebook, VS Code

WORK EXPERIENCE

AI/ML Engineering Intern | KALNET GLOBAL

- Built a multi-agent RAG assistant using Python, LangGraph, BGE embeddings, and Qdrant for domain-specific knowledge retrieval.
- Developed ingestion and chunking pipelines for Markdown, text, PDF, and DOCX documents using 384-dimensional vector embeddings.
- Implemented semantic routing, answerability checks, and context-grounded generation to safely handle unsupported queries.
- Created an evaluation framework covering retrieval, routing, groundedness, faithfulness, and adversarial scenarios.

Outlier AI - AI Trainer (Freelancer)

- Evaluated **200+ LLM-generated C++/Python and mathematical reasoning responses** over **100+ hours**, identifying hallucinations, logical inconsistencies, reasoning gaps, and implementation errors to improve response quality and model reliability.
- Applied **structured evaluation criteria** to analyze recurring errors and provided **evidence-based feedback** to improve AI-generated technical outputs and training data quality.

PROJECTS

Talk-to-Your-Data

- Developed a Gemini-powered text-to-SQL pipeline that passed **all 25 cases in a curated analytical benchmark**.

- Improved SQL generation through schema-grounded prompts, real station values, and automated error-feedback correction.
- Built a resumable evaluation framework to measure answer correctness, latency, and provider failures.
- Added SQL safety guardrails and graceful AI fallbacks to make LLM responses more reliable and transparent.

TripMate AI--Multi-Agent Travel Planning System--Live Demo

- Architected an AI travel planner using FastAPI and LangGraph, orchestrating 4 specialised agents connected to external APIs via Model Context Protocol (MCP).
- Optimised LLM token consumption by replacing model-based entity extraction with deterministic parsing, reducing the workflow to exactly 2 LLM calls per request and eliminating rate limits.
- Engineered a fault-tolerant PostgreSQL backend and established a Docker-containerised CI/CD pipeline via GitHub Actions, automating 25+ tests to enforce strict code coverage.

Bank Customer Segmentation & Predictive Churn Modeling

- Optimized K-Means customer segmentation, improving the Silhouette Score by 65% (0.11 0.18) by prioritizing behavioral features over noisy demographic variables.
- Validated cluster stability across 25 bootstrap resamples, achieving a mean Adjusted Rand Index (ARI) of 0.85 using PCA and stability diagnostics.
- Established statistical significance of customer segments and churn risk using ² = 309.3, p < 0.001, Cramér's V = 0.175, with 95% confidence intervals for cluster-level churn rates.
- Built a Random Forest churn prediction baseline, achieving 0.85 ROC-AUC and 0.69 precision, separating exploratory segmentation from supervised classification.

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures & Applications, Operating Systems
- **Mathematics:** Linear Algebra, Probability, Statistics, Calculus, Differential Equations, Complex Variables
- **Machine Learning:** Fundamentals of Machine Learning

POSITIONS OF RESPONSIBILITY

Head Internship Coordinator | Computational Engineering, IIT Hyderabad

- Led a 5-member team managing internships for 30 students, coordinating 150+ interviews, 20+ assessments, and outreach to 100+ companies.

Placement Coordinator | Office of Career Services, IIT Hyderabad

- Pitched to **100+ companies and converted 20** for IIT Hyderabad's placement drive; served as company POC during the **2-day Japan Career Day.**

EXTRACURRICULAR

- Gained first-hand exposure to 21 Japanese companies and Japanese work culture through industry interactions and peers with internship experience in Japan, strengthening my interest in building a career in Japan.
- Active National Service Scheme(NSS) volunteer, participating in various social events and activities on and off campus.
- Enjoy playing badminton and cricket, and I make time for them whenever I can.